Cardiovascular System Modeling Using 0-Dimensional Models

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Abstract

0-Dimensional modeling is considered for representing cardiovascular system functioning. The Grodins model for the cardiac system as a feedback regulator is presented and its development and use cases are explored. In so doing the cardiovascular system is broken up into four compartments: two pumps representing the left and right ventricles, a pulmonary circuit, and a systemic circuit. The pumps and pulmonary circuit are combined to arrive at the Starling Heart-Lung Preparation and the systemic circuit is incorporated to close the system. Analysis of the full system makes up a set of 23 equations, 23 unknowns, and 14 independent variables. These are varied to represent different physiological phenomena including myocardial weakness. Grodins's method of validating the results of his model is briefly explained. It is determined that the model is sufficient for acting as an accurate feedback regulator model for the cardiovascular system. The model's functions in comparison to the Windkessel model are discussed and contributions that build on the Grodins model are presented.

Keywords: Grodins, Cardiovascular, 0-Dimensional, Model, Feedback Regulator

1 Introduction

The cardiovascular system consists of the heart with a left and right ventricle, the systemic arteries which carry oxygenated blood from the heart to the rest of the body, and the pulmonary vessels, which carry blood through the lung for oxygenation. Zero-dimensional models provide a concise way to model this complex system. Using a 0-D model, the circulatory system can be split into different compartments and each analyzed to create a model describing their function. These models assume a uniform distribution of pressure, flow, and volume within any compartment at a specific instant in time. 0-D models do not describe variations in pressure or flow with spatial position.

Various models of the cardiovascular system have been developed by different researchers[1]. The first 0-D cardiovascular model was the two-element Windkessel model proposed by Stephen Hales in 1733 and formulated by Otto Frank in 1899[4]. From the 1940s to the 1950s, several other models were developed using the Windkessel model as a basis. In 1959 model, pioneer Fred Grodins developed a cardiovascular model based on control theory. The purpose of this

paper is to explore this model to understand cardiac hemodynamics through pressure regulation modeling.

In an attempt to delve into the Grodins cardiac model, we will consider the motivation, development, and accuracy of the model. The model will be compared to the Windkessel model as it was presented in this course.

2 Methodology

Motivation behind the Grodins model originates from controversies about cardiac output regulation during the 1940's and 50's. The two most disputed views of cardiovascular control were 1) that the heart controls cardiac output and 2) that vascular resistance controls blood pressure[3]. Grodins believed there was a need for basic integration of cardiovascular dynamics, thus creating his 0-D model of the cardiovascular system.

The Grodins model focuses on simulating steady-state conditions of the cardiovascular system by representing the system as a feedback regulator. In the feedback regulator representation, a controlling system passes quantities to the controlled system, which in turn feeds back controlled quantities to the controlling system. This phenomenon is demonstrated in Figure 1.

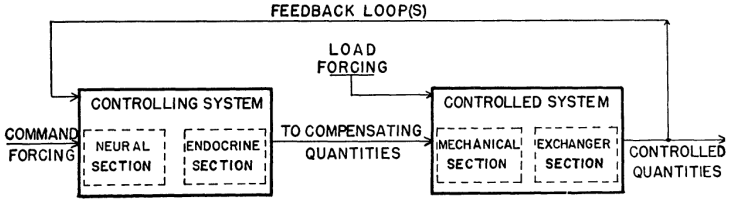
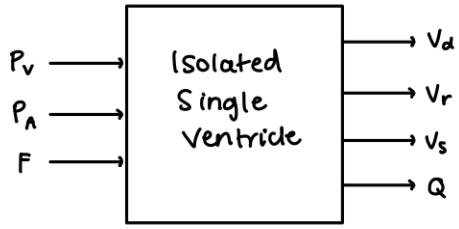


FIG. 1. THE CARDIOVASCULAR REGULATOR

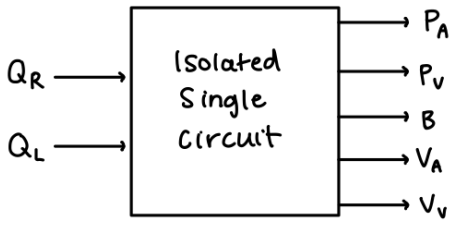
Figure 1: Feedback Regulator

Grodins split the cardiovascular system into four isolated compartments: two pumps for the left and right ventricles, a circuit for pulmonary function, and a circuit for the function of the systemic vessels. The approach involves isolating each component, identifying the inputs and outputs of each, and analyzing the interactions between these inputs and outputs in order to develop a set of mathematical equations that help describe cardiac function. Developing the model first requires analysis of each isolated component. Consider the isolated ventricle shown in Figure 2a. The inputs to the isolated ventricle are venous pressure (P_V), arterial pressure (P_A), and cardiac frequency (F). The outputs of the isolated ventricle are then the diastolic ventricular volume (V_d), stroke ventricular volume (V_s), residual ventricular volume (V_r), and cardiac output (Q). Similarly, consider the isolated circuit shown in Figure 2b, applicable to either of the pulmonary or systemic systems. Inputs to the isolated circuit include the cardiac outputs from the left and right pumps. Outputs from the isolated circuit are the venous and arterial pressures and volumes (P_V, P_A, V_V, and V_A), and the total blood volume (B). An important thing to note here is the use of an "open volume" circuit. The circuit presented in Figure 2b uses the concept that one pump transfers blood from an infinite external blood source to the central artery and the other transfers blood from the

central vein to an external drain. In this model, we have transient inequalities between Q_R and Q_L such that the total blood volume becomes dependent instead of independent.



(a) Isolated Ventricle Block Diagram



(b) Isolated Circuit Block Diagram

Figure 2: Isolated Compartments

The next step in developing Grodins's model is to present the Starling Heart-Lung Preparation. This preparation incorporates function of the heart with that of the pulmonary system, as shown in Figure 3. The left and right ventricular pumps are connected with the pulmonary circuit. In this case, the independent variables are the venous and arterial systemic pressures (P_{V_S} and P_{A_S}) and the cardiac frequency, F. The cardiac outputs from the left and right ventricles are fed into the pulmonary circuit, and the venous and arterial pulmonary pressures (P_{V_P} and P_{A_P}) are fed back into each pump.

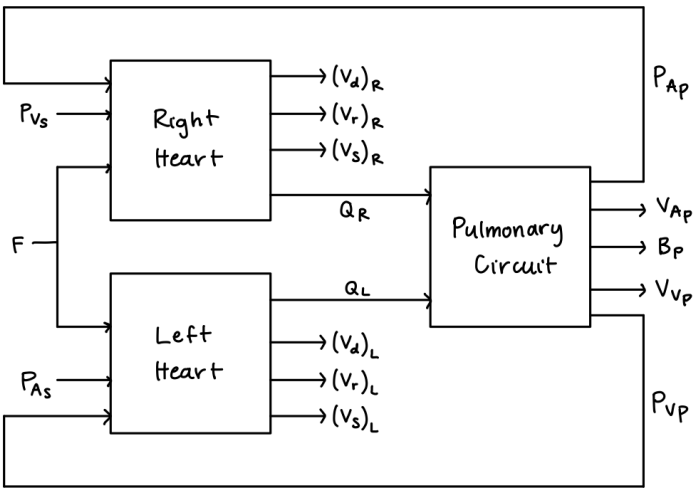


Figure 3: Starling Heart-Lung Preparation

To obtain the full mechanical model shown in Figure 4, the systemic circuit is connected to the pulmonary circuit in series. The addition of the systemic circuit closes the loop of the feedback regulator, and now we have a closed system. The only independent variable to the regulator is cardiac frequency. All venous and arterial pressures are fed back to the ventricle pumps from the circuit compartment. Obtaining the mathematical equations to describe cardiac function require finding the other independent variables of the system. These will come from the inner functions

of the left pump, right pump, and pulmonary and systemic circuit compartments. There will be 6 additional variables from the cardiac properties: left and right ventricle strengths (S_L, S_R), left and right ventricular compliance (C_L, C_R), and left and right resistance to ventricular filling (R_L, R_R). There will also be an additional 7 variables from the circuit properties consisting of pulmonary and systemic arteriole resistances (R_P, R_S), arterial and venous pulmonary and systemic complinaces (C_{A_P}, C_{V_P}, C_{A_S}, C_{V_S}), and total blood volume (B).

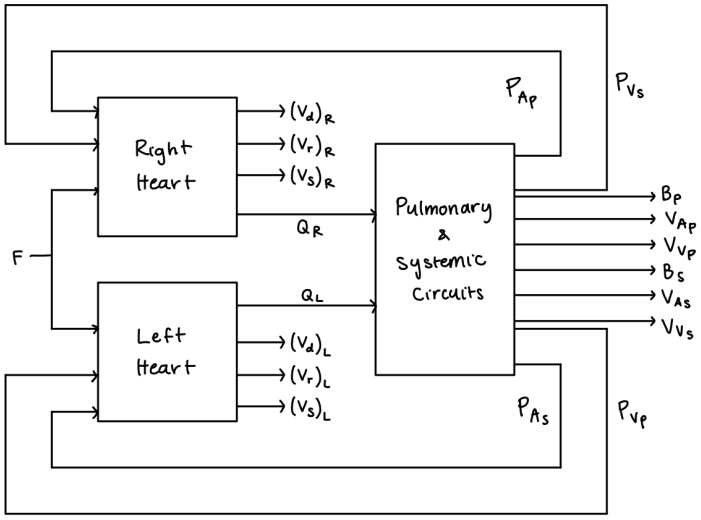


Figure 4: Full Mechanical System

To develop the driving equations, we begin with examining the steady-state emptying and filling of the isolated ventricle. During an isolated ventricular systole, the ventricle starts with a full diastolic volume, V_d. The ventricle ejects a stroke volume, V_s, and a residual volume, V_r, is left over. We define a 'stroke work':

$$w_s = S \cdot V_d \tag{1}$$

Then we can find the dependence of V_s and V_r on V_d and arterial pressure, P_A. Doing so leads to the following relations for ventricular filling:

$$V_s = \left(\frac{S}{V_d}\right) V_d \tag{2}$$

$$V_r = \left[1 - \frac{S}{V_d}\right] V_d \tag{3}$$

Now consider an isolated ventricular diastole, where the ventricle begins with volume V_r and fills for a time t to volume V_d. A number of assumptions are made to arrive at the filling law: 1) ventricular relaxation occurs instantaneously, 2) filling of the relaxed ventricle is a first order linear process, and 3) the unstressed volume of the relaxed ventricle is zero. These assumptions ultimately lead to an ODE filling law:

$$RC\dot{V}_d + V_d = CP_V \tag{4}$$

Using the condition that $V_d = V_r$ at time $t = 0$ yields the following solution:

$$V_d = CP_V + (V_r - CP_V)e^{-t/RC} \tag{5}$$

Manipulating equations and defining cardiac frequency parameters A and K, gives the following relationship for cardiac output:

$$Q = \frac{FSCAP_V}{AP_A} + SK \tag{6}$$

Following a similar approach gives a set of equations for the systemic and pulmonary circuits. These are given in Table 1.

Table 1: Equations for the Systemic and Pulmonary Circuits

Systemic	Pulmonary
$F_S = \frac{(P_{A_S} - P_{V_S})}{R_S}$	$F_S = \frac{(P_{A_P} - P_{V_P})}{R_P}$
$P_{A_S} = (1/C_{A_S})V_{A_S}$	$P_{A_P} = (1/C_{A_P})V_{A_P}$
$P_{V_S} = (1/C_{V_S})V_{V_S}$	$P_{V_P} = (1/C_{V_P})V_{V_P}$
$B_S = V_{A_S} + V_{V_S}$	$B_P = V_{A_P} + V_{V_P}$

The full system consists of 23 equations, 23 unknowns, and 14 independent variables. The development of this model allows the user to vary any single or combination of these independent variables and determine how the other cardiovascular properties change.

3 Findings

Grodins's idea for using this model was that the user could change input variables to model different physiological phenomenon. As there are 14 independent variables, there are 4,782,954 different possible simulations to run. His approach to validating this model was to run various examples representing physiological phenomenon and compare them to either experimental data or other validated models created by other researchers in the field. An example we present here is his simulation of myocardial weakness. Myocardial weakness occurs when the strength of the ventricle is reduced. Figure 5 shows the results Grodins obtained by decreasing the strength of the left and right ventricles (S_L and S_R)[2]. With the weakening of one or the other ventricles, cardiac output for that ventricle decreases and blood accumulates in the pump behind the weakened ventricle. Increasing the blood volume in the system helps to restore the cardiac output. Grodins compared this result to another mechanical model developed by colleagues and found that the two were in great agreement. Grodins also tested phenomenon like higher cardiac frequency, increased blood volume, and increases peripheral resistance. He found success in these models as well.

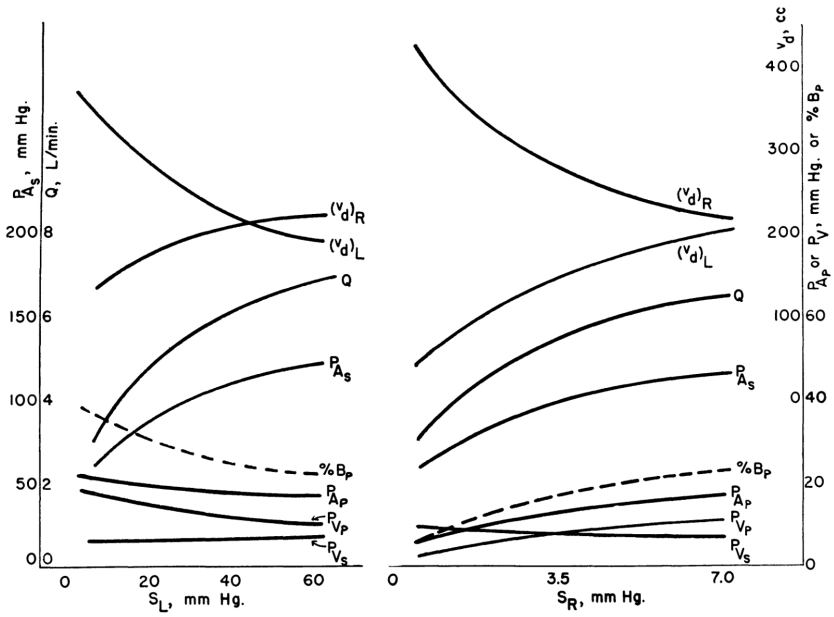


FIG. 20. THEORETICAL RESPONSES, LEFT, TO REDUCTION IN S_L, AND RIGHT, TO REDUCTION IN S_R

Figure 5: Myocardial Weakness Simulation

4 Discussion

The goal of our research was to understand different 0-dimensional cardiovascular system models. By focusing our research on the Grodins Model, we were able to understand the importance of adding pressure regulation and multi-compartment models to the simpler Windkessel model.

The overall circuitry is based on electrical elements for lumped-parameter modeling. The 2-element Windkessel model uses compliance and resistance to model the entire cardiovascular system. Grodins's model uses a unique compliance and resistance in each one of its 4-compartments. The 3-element Windkessel model uses characteristic impedance to be to accurately represent systolic pressure and the 4-element Windkessel model uses total arterial inertance to represent high frequencies in the model [5].

Even though the 3 and 4-element models have a greater number of circuity elements, the Windkessel model is still unable to represent local vascular changes or activity of the more distal flow from the heart due to its single compartment design [5]. Grodins's 4-compartment design is capable of greater accuracy due to its unique resistances and compliances for each compartment.

The Grodins model is capable of modeling specific steady state scenarios of the cardiovascular system through the 14 independent variables. Using these inputs, a number of scenarios are capable of being modeled. These scenarios provide a unique opportunity to model different phenomena and steady state scenarios. Varying the 14 independent variables allows the understanding of interrelationships of the cardiovascular system. The Grodins model allows for the ability to see how dependent variables are affected in these scenarios. This capability is not available through

previous models, specifically the Windkessel models.

Grodins's model has the ability to model pressure regulation present in the cardiovascular system through the baroreflex mechanism. The baroreceptors in the cardiovascular system allow the body to regulate blood pressure through heart rate. When blood pressure increases, a signal is sent to decrease heart rate, and vice versa. This pressure regulating mechanism is model by Grodins's by control theory, but is not present in Windkessel's models.

The concepts from Grodins model were used by many future scientists to model the cardiovascular system. These models include Kappel and Peer who used the Grodins Model but added in a workload input. This model was intended to represent the cardiovascular system during exercise [1]. In addition, Batzel used inspiration from both the Grodins Model and Kappel and Peer's model to create a system that regulates both heart rate and pressure through a single feedback loop. The models based off of Grodins use of control theory emphasize that his comprehensive design is still influencing models today.

5 Conclusion

Lumped Parameter modeling provides the unique opportunity of modeling the cardiovascular system using circuit elements. Although the 0-dimensional elements are unable to model wave phenomena in cardiovascular flow, they have the capabilities to provide cheap calculations for understanding this complex system. The comprehensive 0-dimensional mode created by Grodins provides a simple yet accurate way to estimate cardiovascular behavior. Grodins's 23 equation and 14 independent variable model introduces interrelationships in the cardiac system that were not previously present in other models. Grodins's model representation of the baroreflex mechanism through control theory and use of a 4-compartment design provides a more accurate representation of the cardiovascular system. Overall, Grodins's development serves as an accurate cardiovascular system model and has set a foundation for further and more complex development of other cardiac models.

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